

Intro to AI for Medicine

Lec 8: Computer Vision and Radiology

Last Time

- Case Studies
 - LSTM
 - Arrhythmia classification
 - Stroke prediction
 - U-Net/CNN -> synthetic MRI

Agenda

- What's different about image data?
- Neuroscientific basis for computer vision – 1981 Nobel Prize
- Intro to Computer Vision
- The Radiologist's workflow
- AI interventions based on different points (only a few!)
- Goals
 - Not teach you everything going on in the world of Radiology
 - Help you fold AI image interpretation into your understanding of AI
 - Brainstorm areas where innovation may be most helpful

Inductive Bias

- Not bias in the traditional sense – it's the direction we point our model toward via the construction of the network
- Ex. For linear regression (no neural net) we assume our data can be fit linearly
- For other problems, we can give our ML model a hint by changing the way it processes data





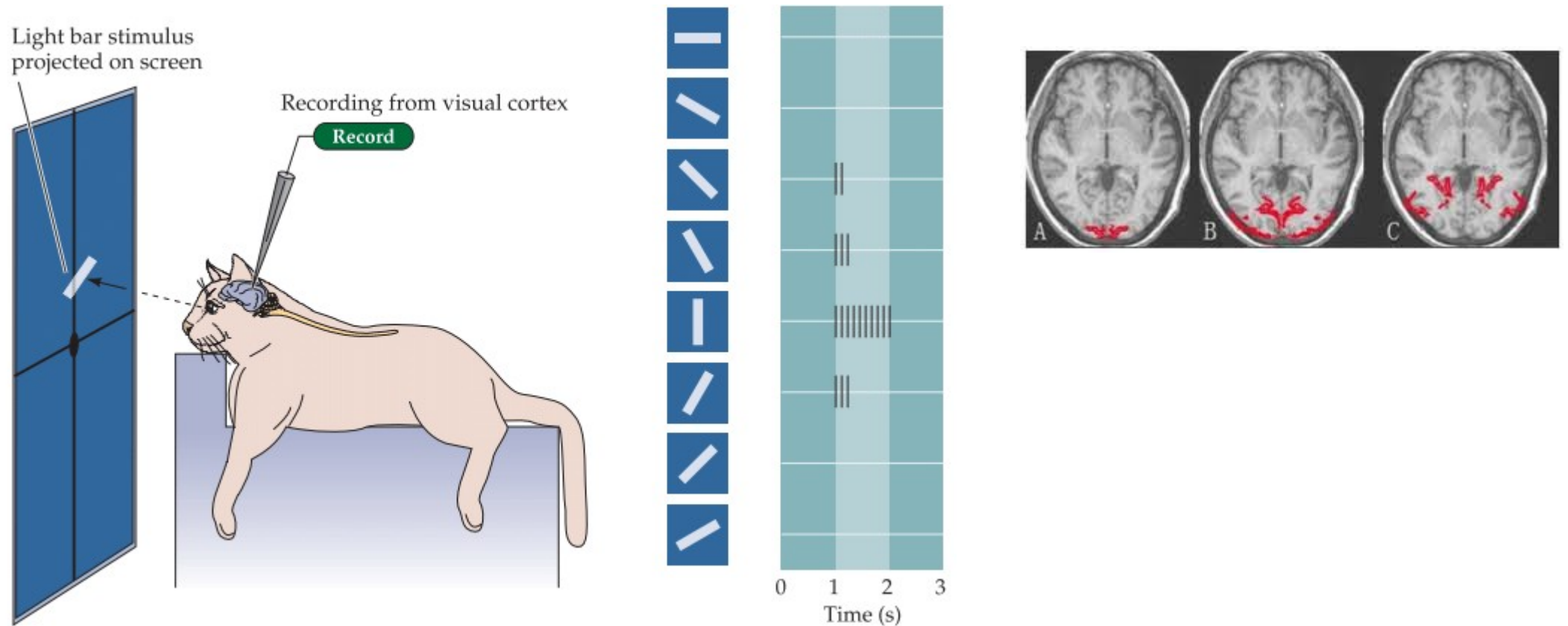


Approaching Imaging With ML



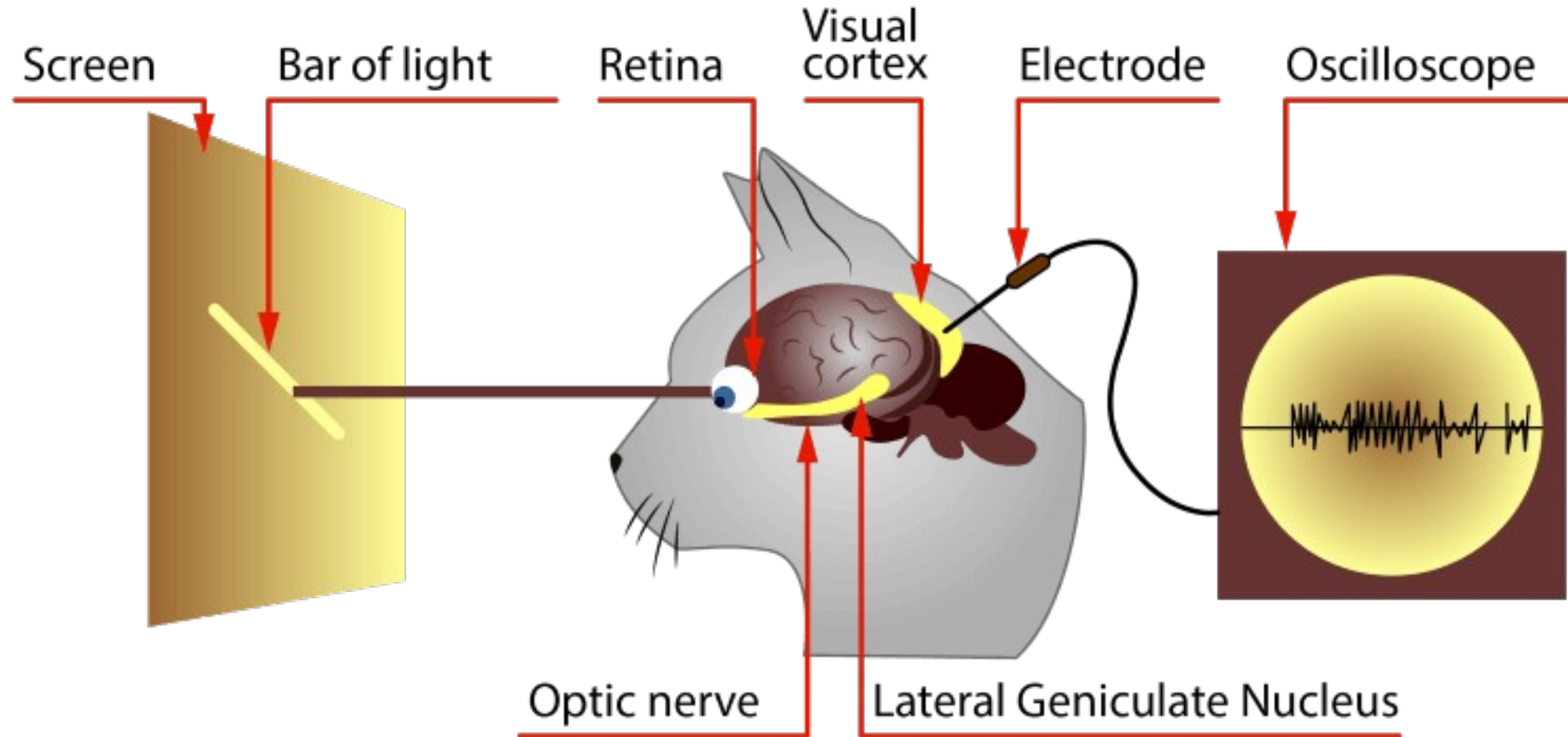
- Recall -> an image is a matrix of BW pixels
- Inductive bias:
 - **Spatial Locality:** Pixels of our target feature will be close together
 - **Translation Invariance:** Pattern of pixels characterizing feature can occur anywhere in image
- Our network should take these into account

Hubel & Wiesel 1981 Nobel Prize

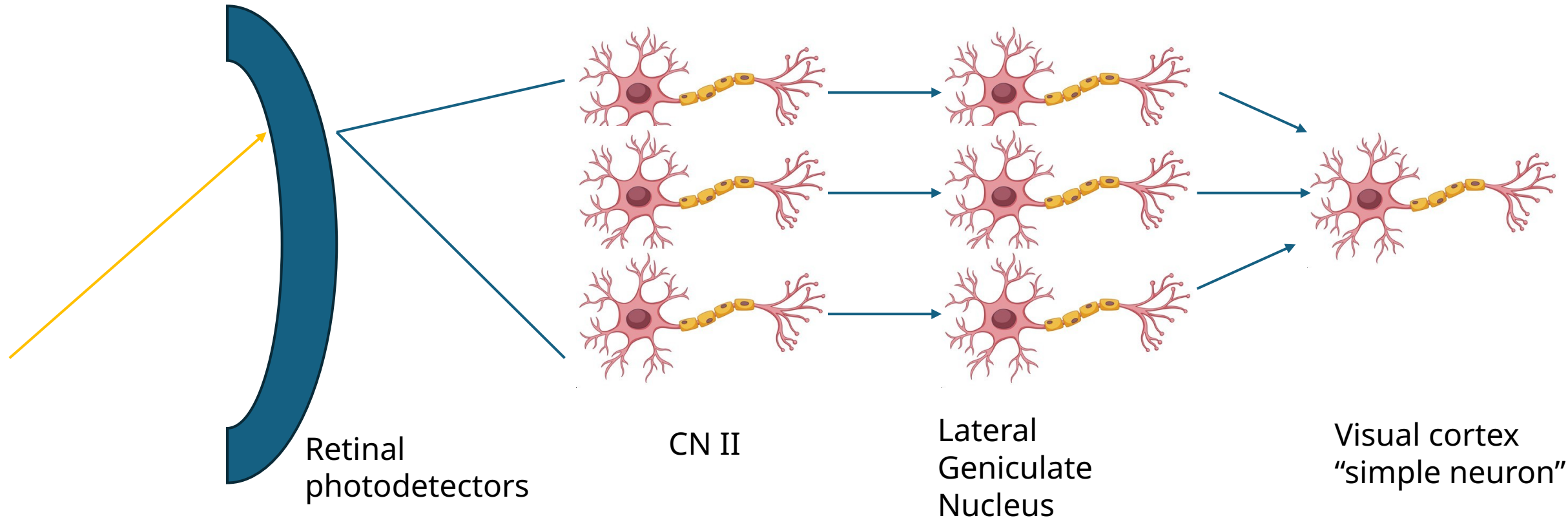


- Anesthetize cat, record from individual neurons @ occipital striate cortex
- Noted increased firing of individual neurons based on light **bar orientations**

Hubel & Wiesel 1981 Nobel Prize

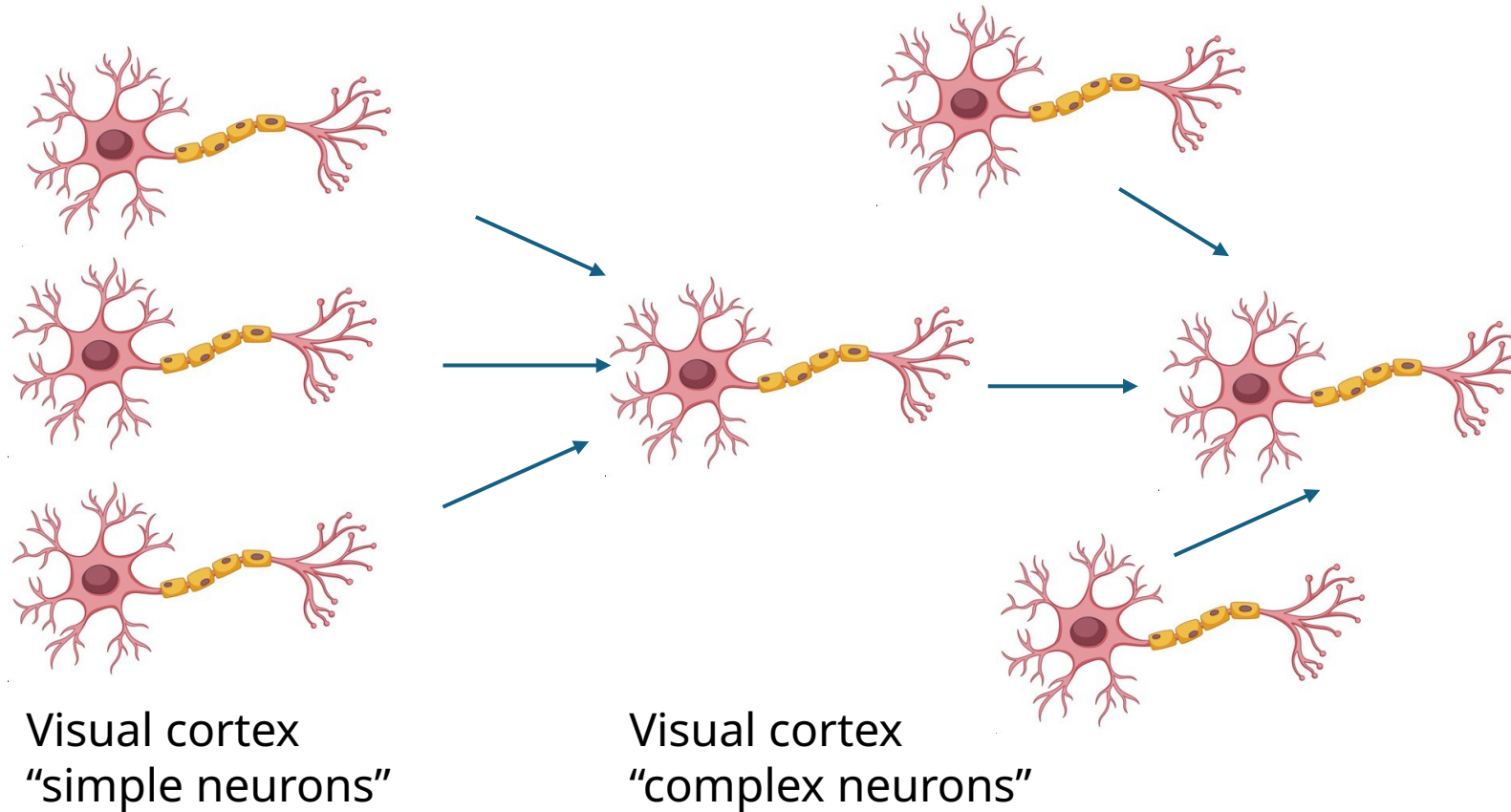


Hubel & Wiesel 1981 Nobel Prize



- Simple neuron synthesizes linear combo of input neurons
- Corresponds to spatial region of retina

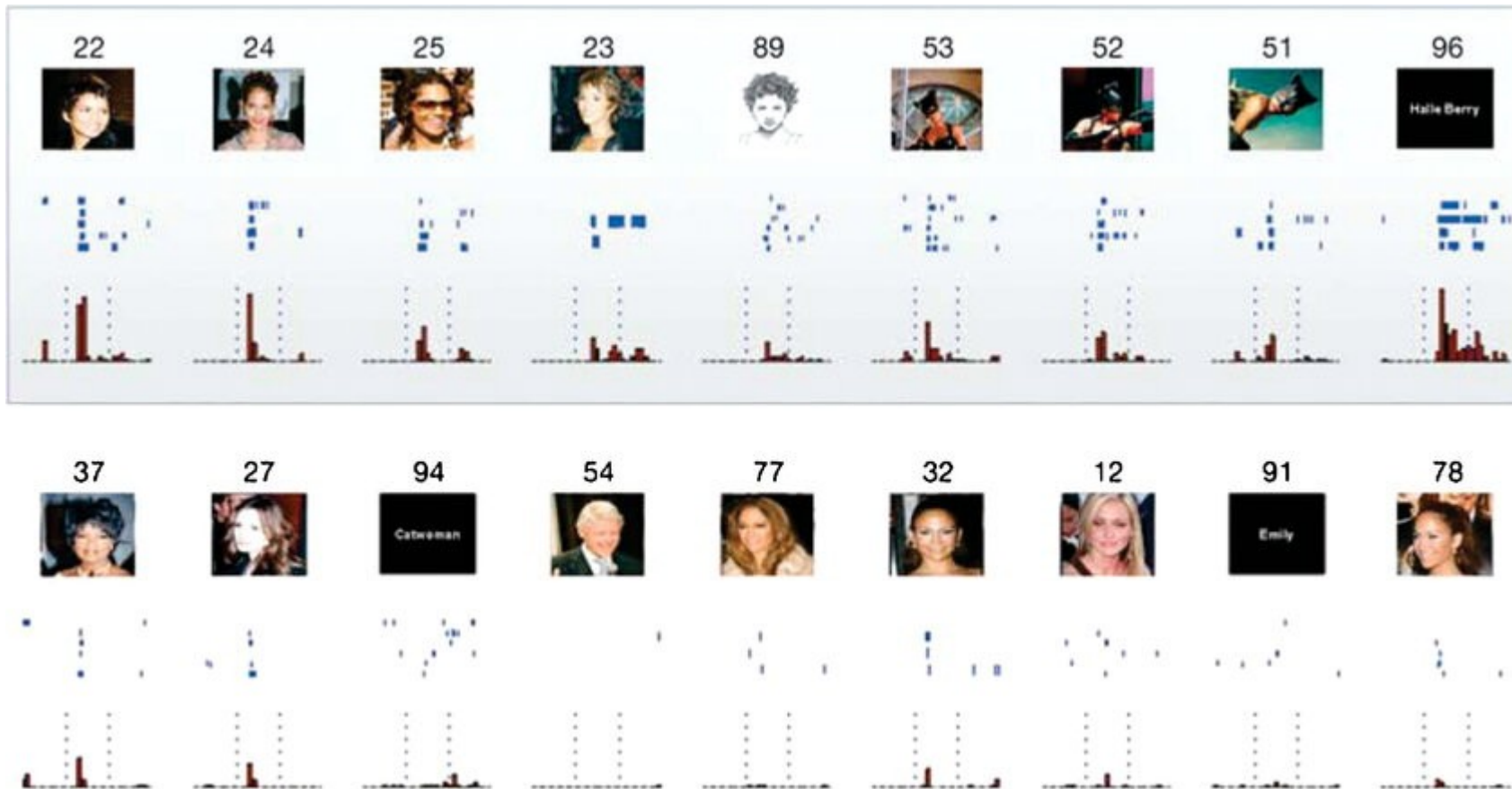
Hubel & Wiesel 1981 Nobel Prize



- Complex neurons: Less predictable effect w.r.t. light intensity and position
- Some noted to pick up certain position of line from any region

Complex Neuron Specialization

- Neuron found that corresponds to Halle Berry – and others



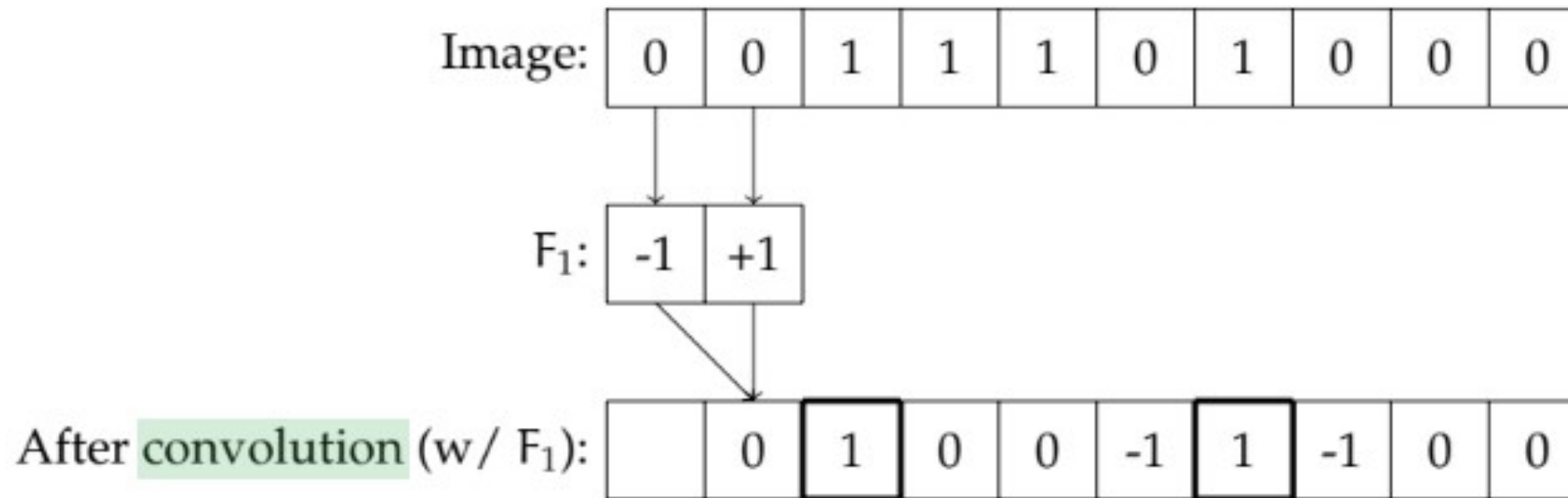
Quiroga, R., Reddy, L., Kreiman, G. *et al.* Invariant visual representation by single neurons in the human brain. *Nature* **435**, 1102–1107 (2005).
<https://doi.org/10.1038/nature03905>

Takeaways

- Two operations going on that we need:
 - **Spatial locality:** Linear combination of spatial region
 - **Translation invariance:** Location irrespective recognition of features

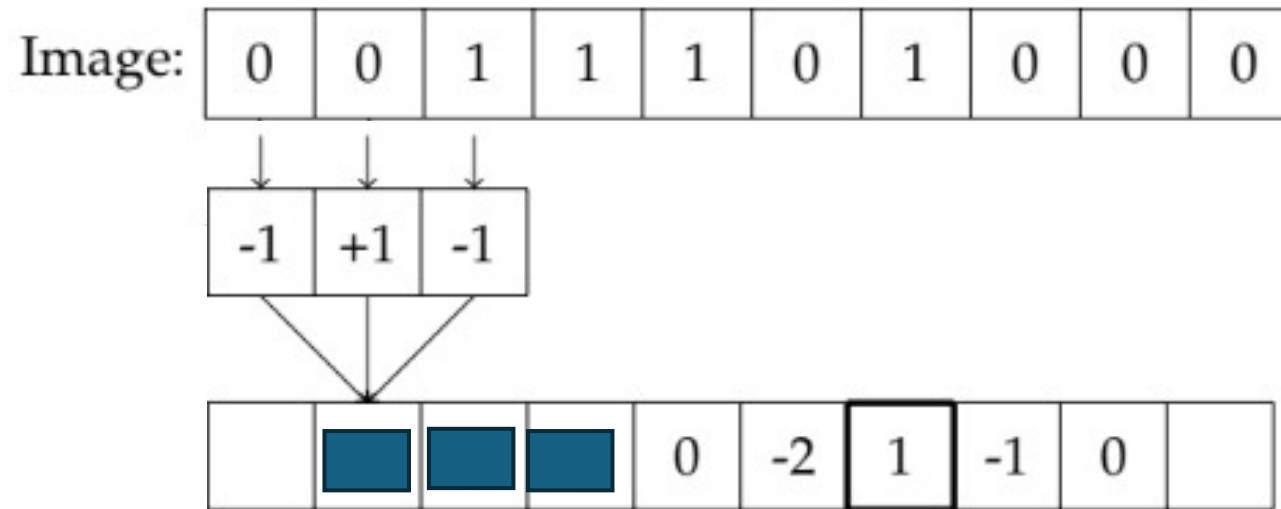
Convolution Operation

- Accounts for **spatial locality**
- Moving matrix (kernel) across the image



Convolution Operation

- Accounts for **spatial locality**
- Moving matrix across the image



Extension to 2D

0	0	0	0	0	0	0
0	60	113	56	139	85	0
0	73	121	54	84	128	0
0	131	99	70	129	127	0
0	80	57	115	69	134	0
0	104	126	123	95	130	0
0	0	0	0	0	0	0

Kernel

0	-1	0
-1	5	-1
0	-1	0

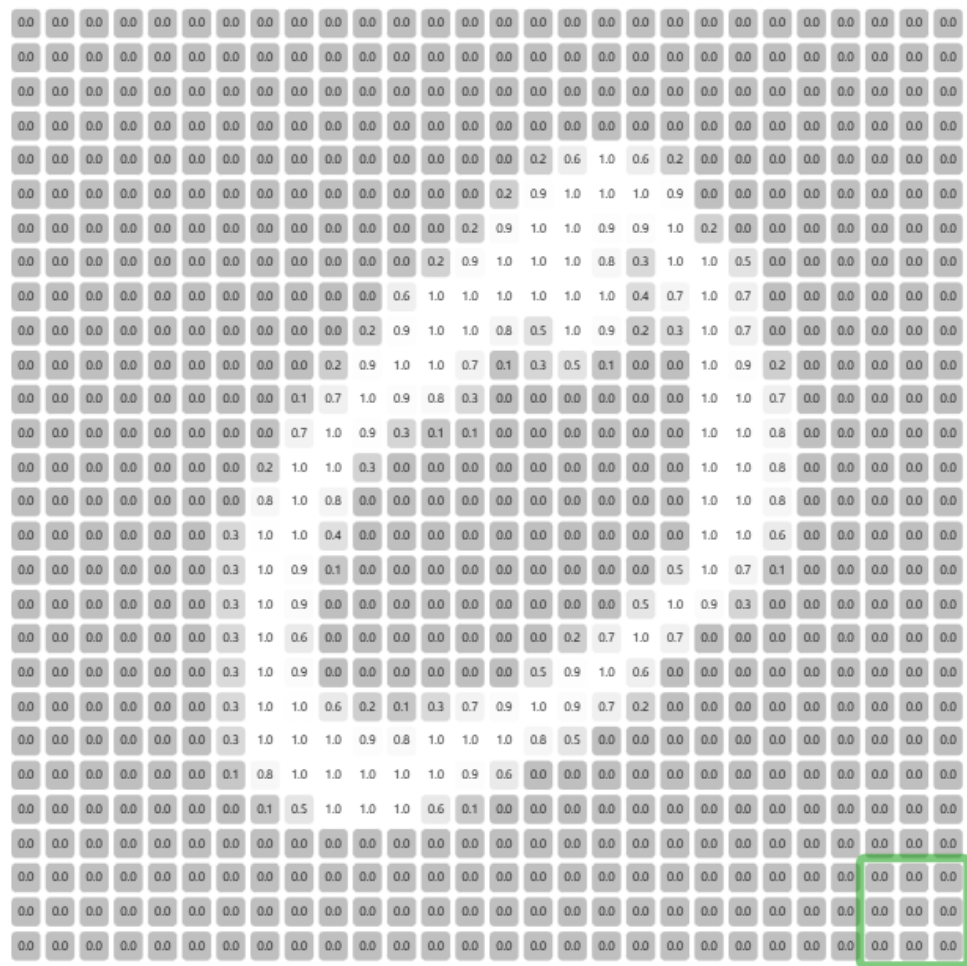
114				

CNN Weights

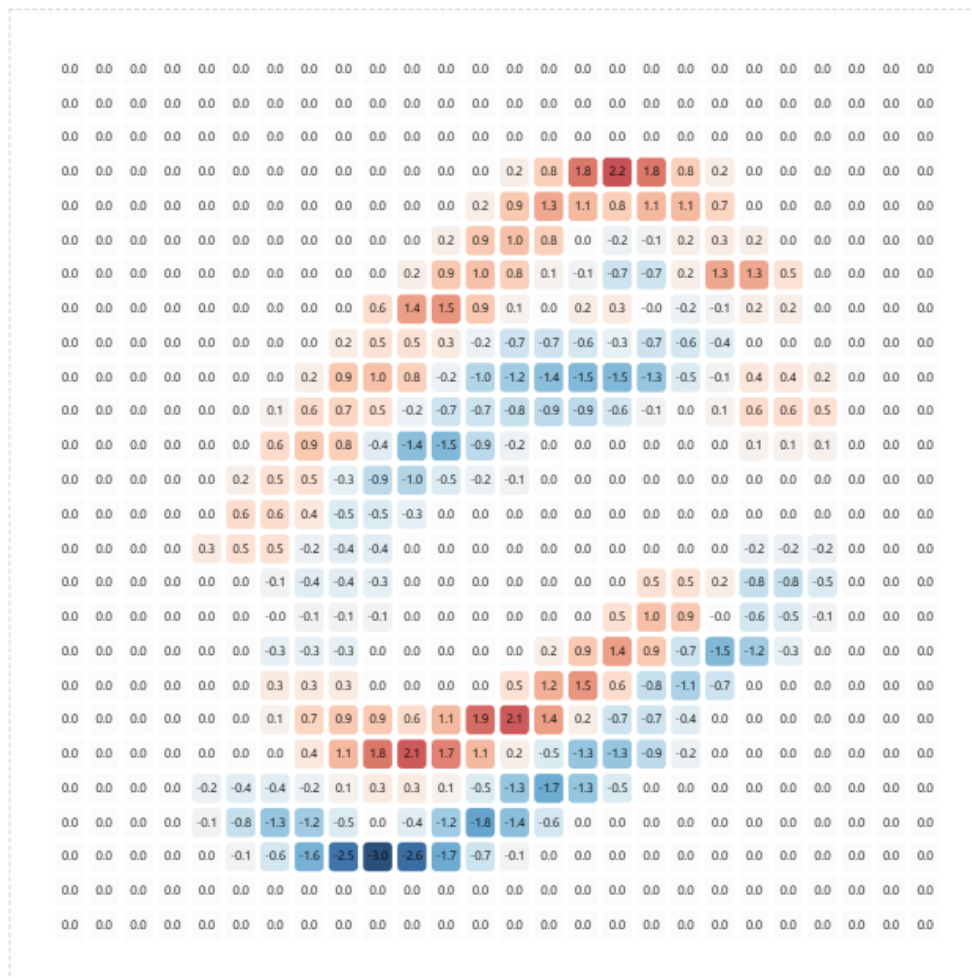
- Improved toward lower loss via gradient descent/backprop
- Shared – every group of pixels gets same kernel
- Each filter learns to extract some critical info from an image.

Types of Filters: Top Edge

Input



Output

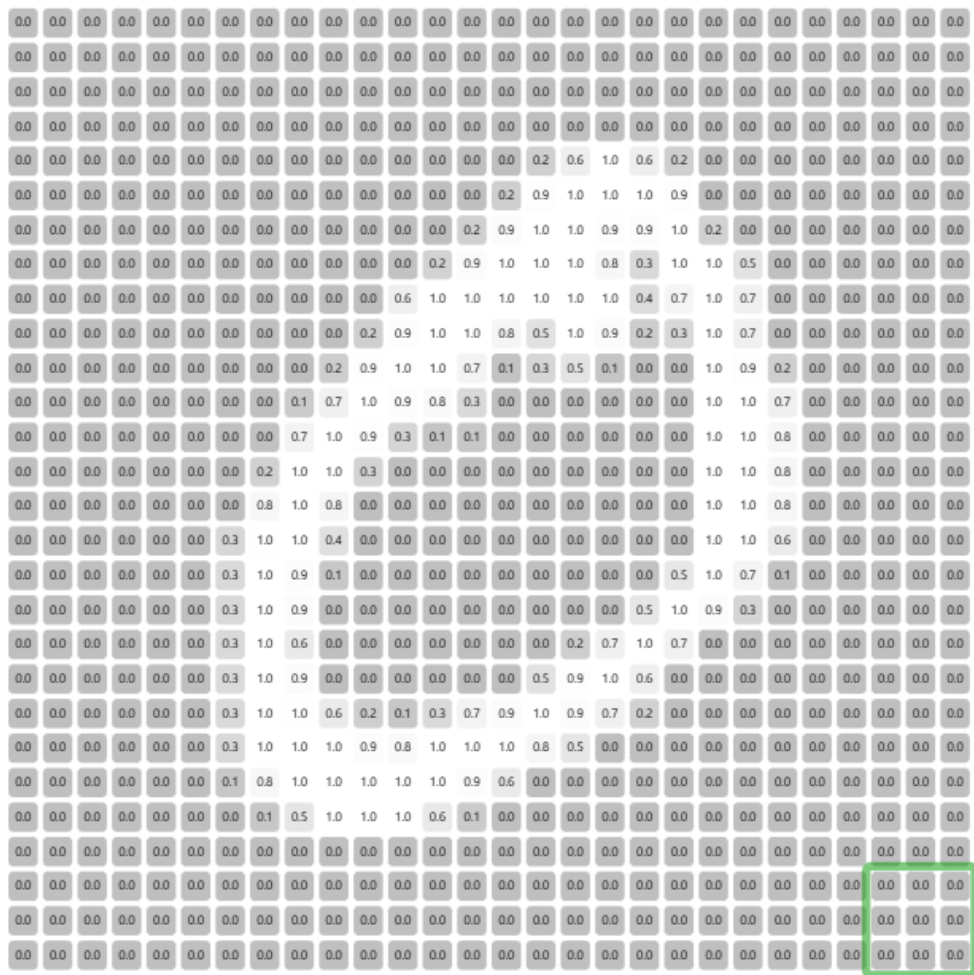


Filter

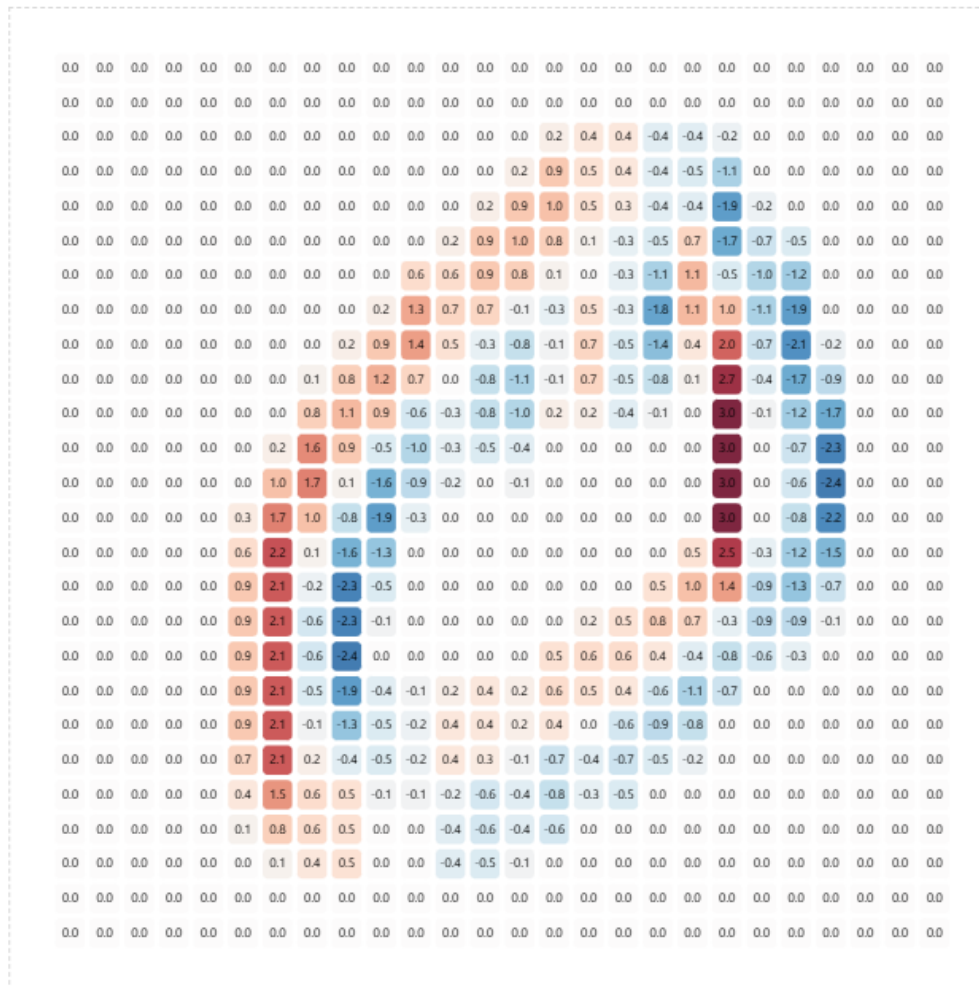


Types of Filters: Left Edge

Input



Output



Filter



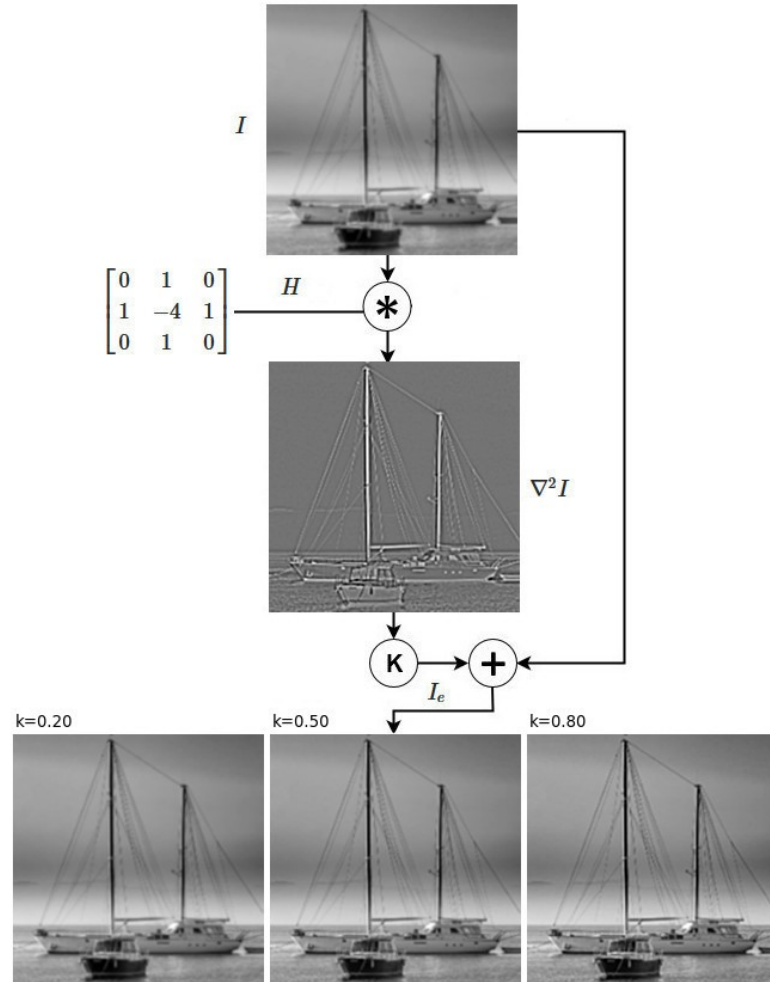
Types of Filters: Smooth



Filter:

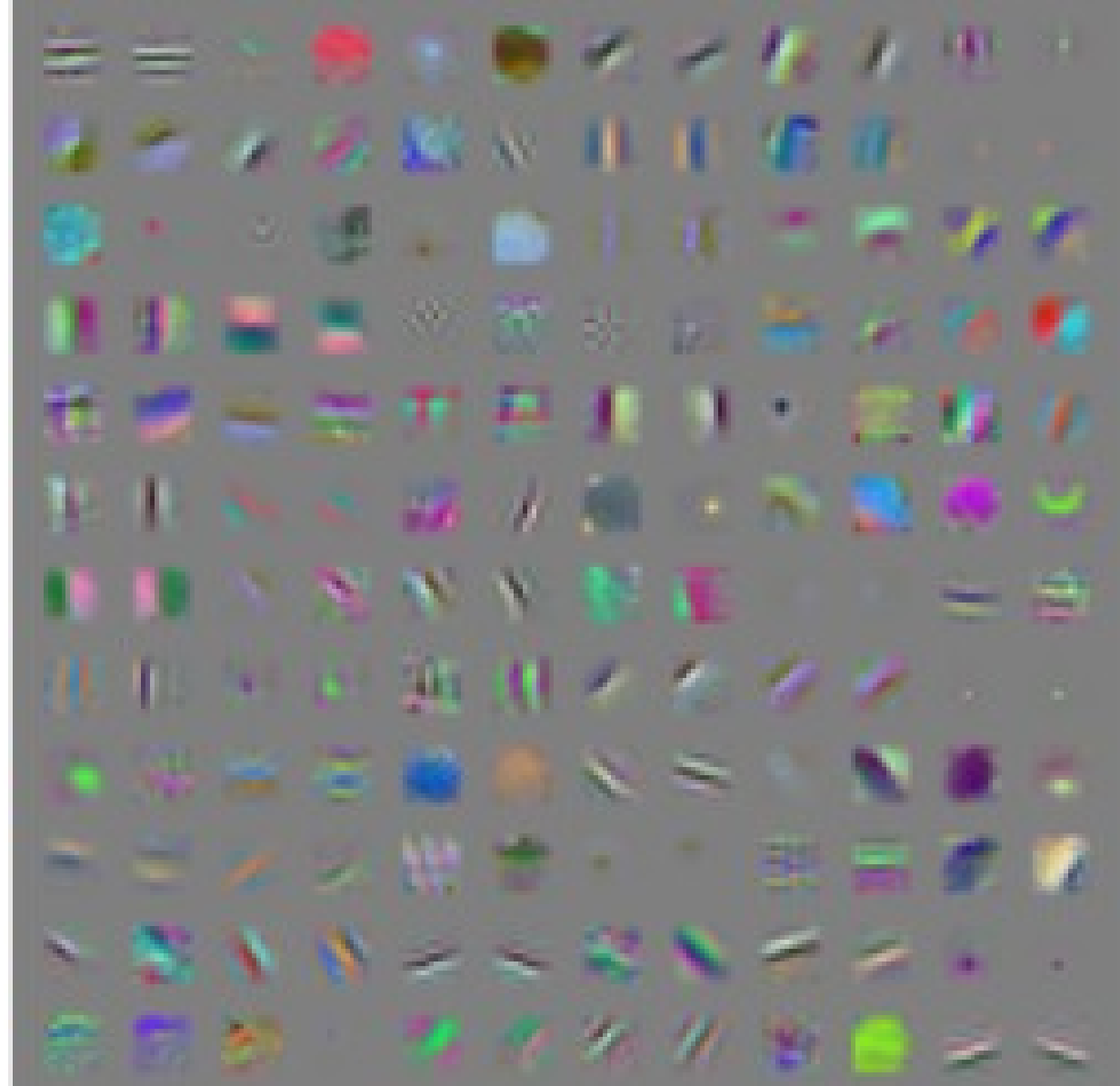
$\begin{bmatrix} 1/9 & 1/9 & 1 \\ 1/9 & 1/9 & 1 \\ 1/9 & 1/9 & 1 \end{bmatrix}$

Types of Filters: Sharpen



$$h_{\nabla} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

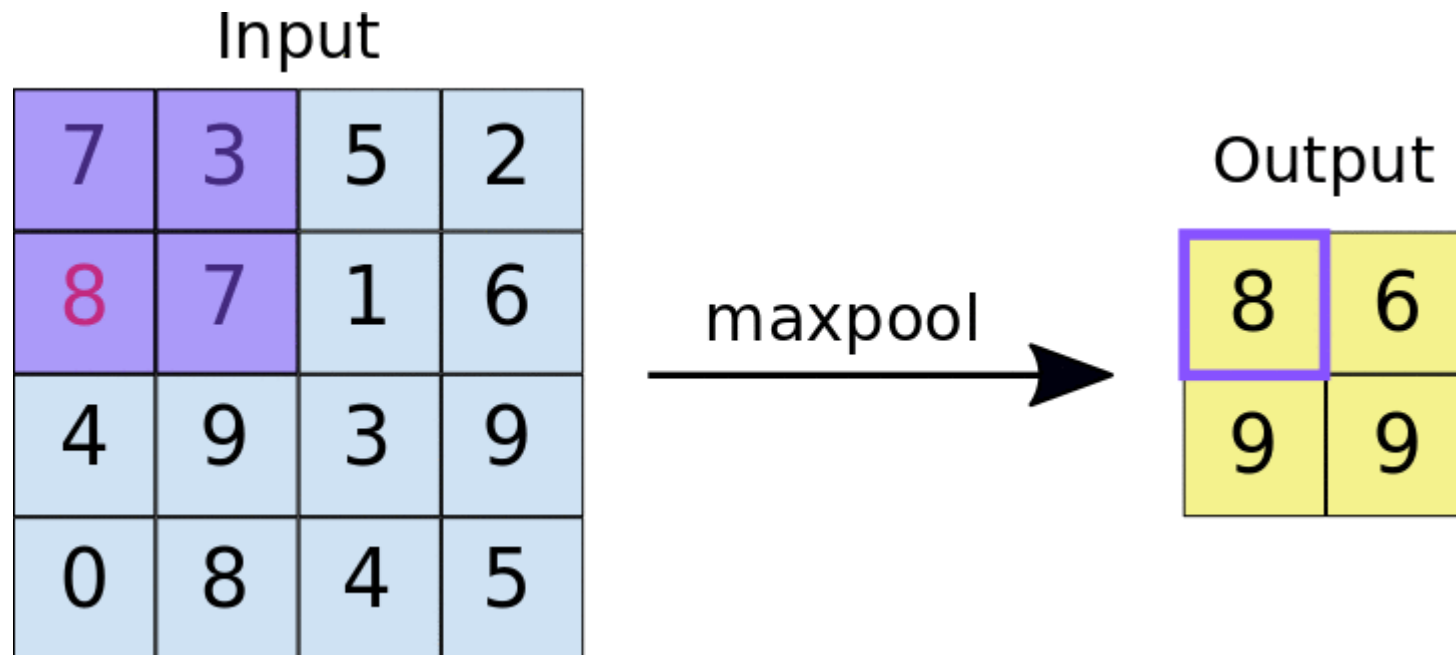
Emphasizes details that are different than surroundings



Convolution kernels learned by the first layer of a fully supervised convolutional maxout network

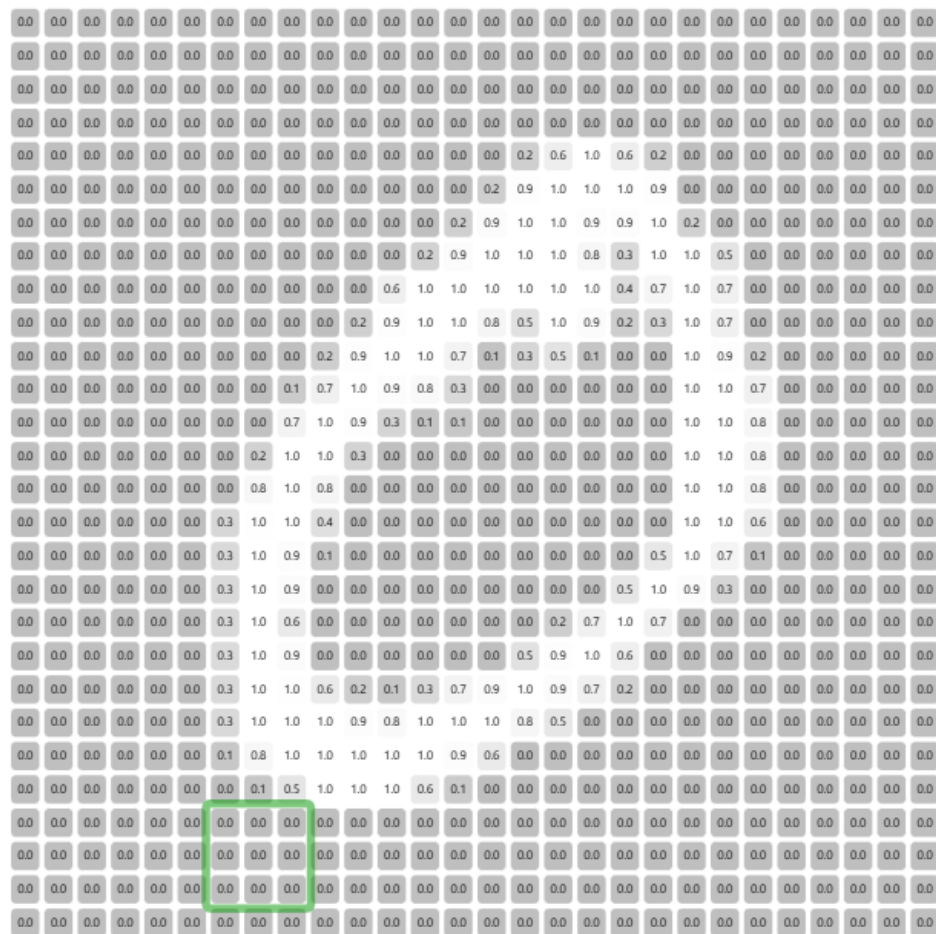
Max Pooling Operation

- Accounts for **Translation Invariance** when combined with convolution
- Takes the maximum value in each field

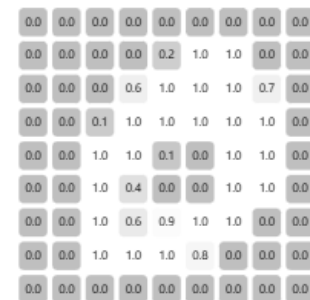


Max Pooling

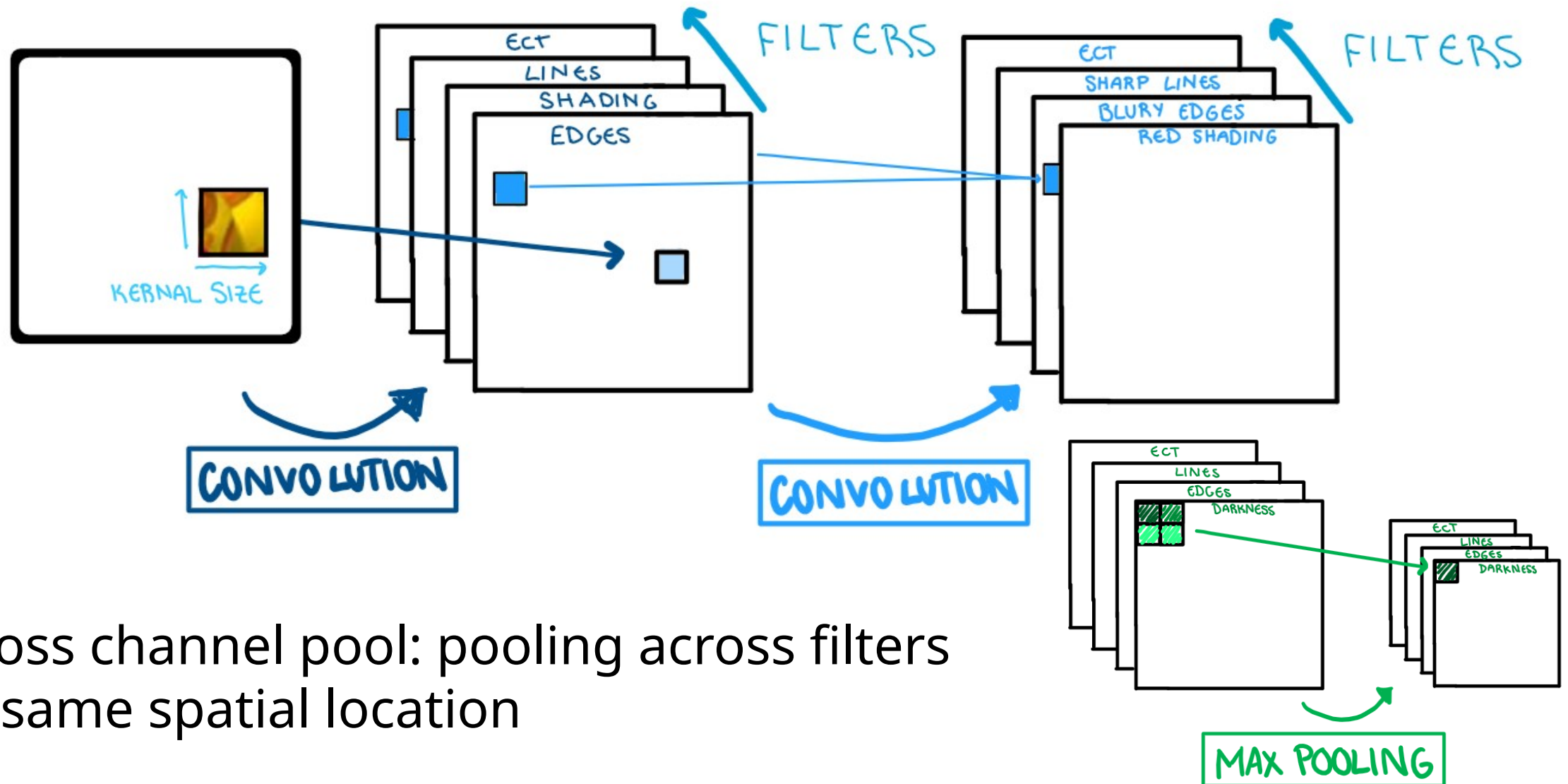
Input



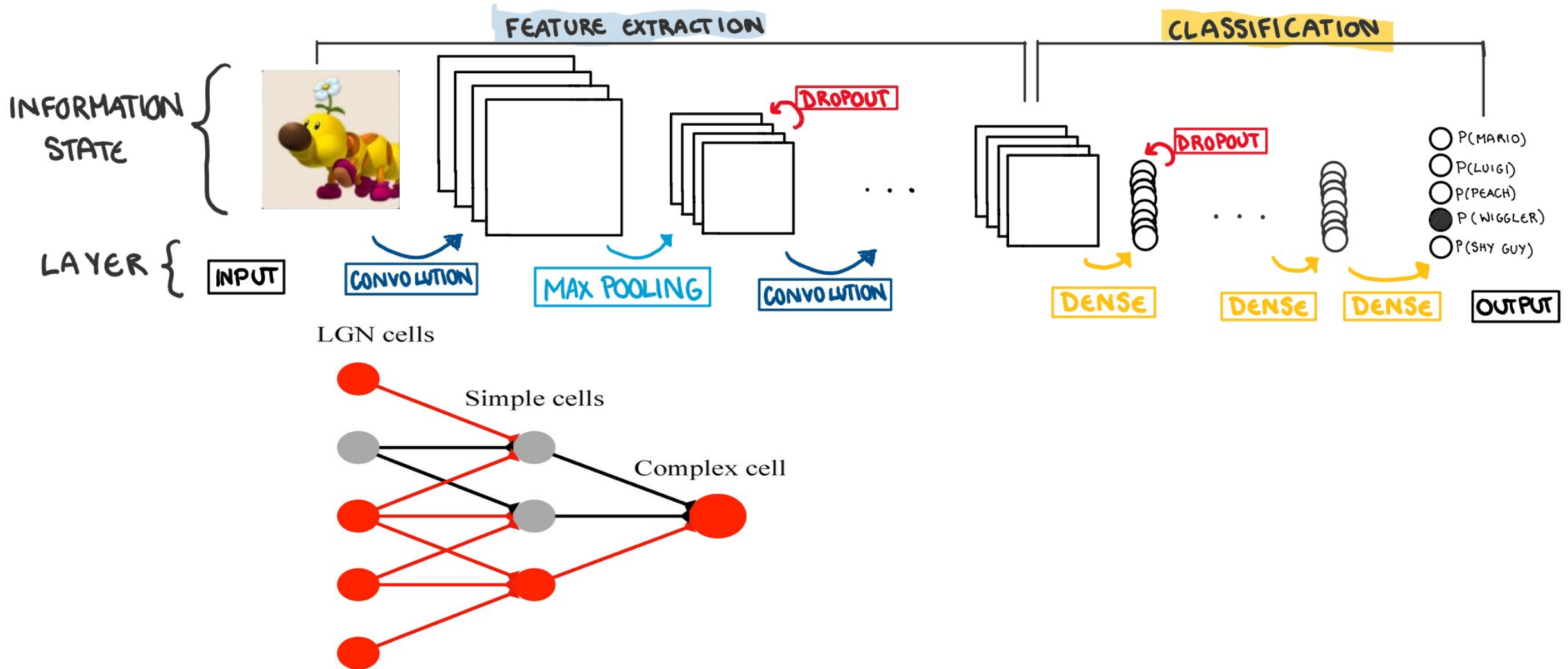
Output



Multi-Filter Parallelization

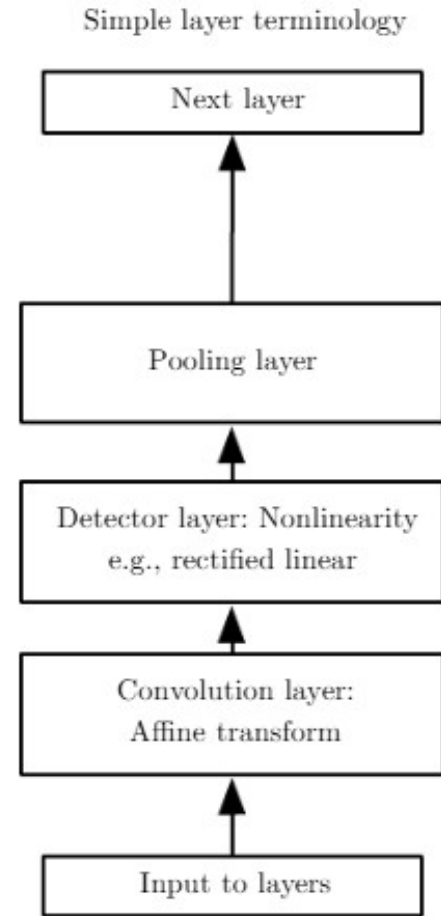
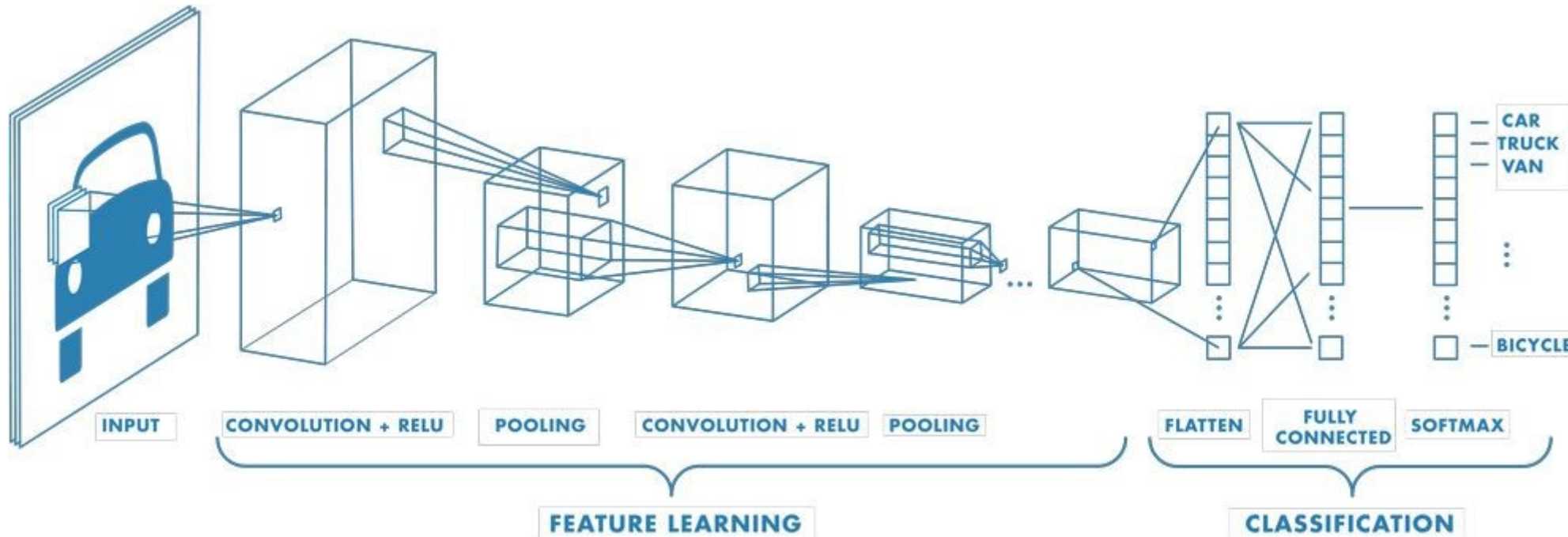


Convolutional Neural Network (CNN)



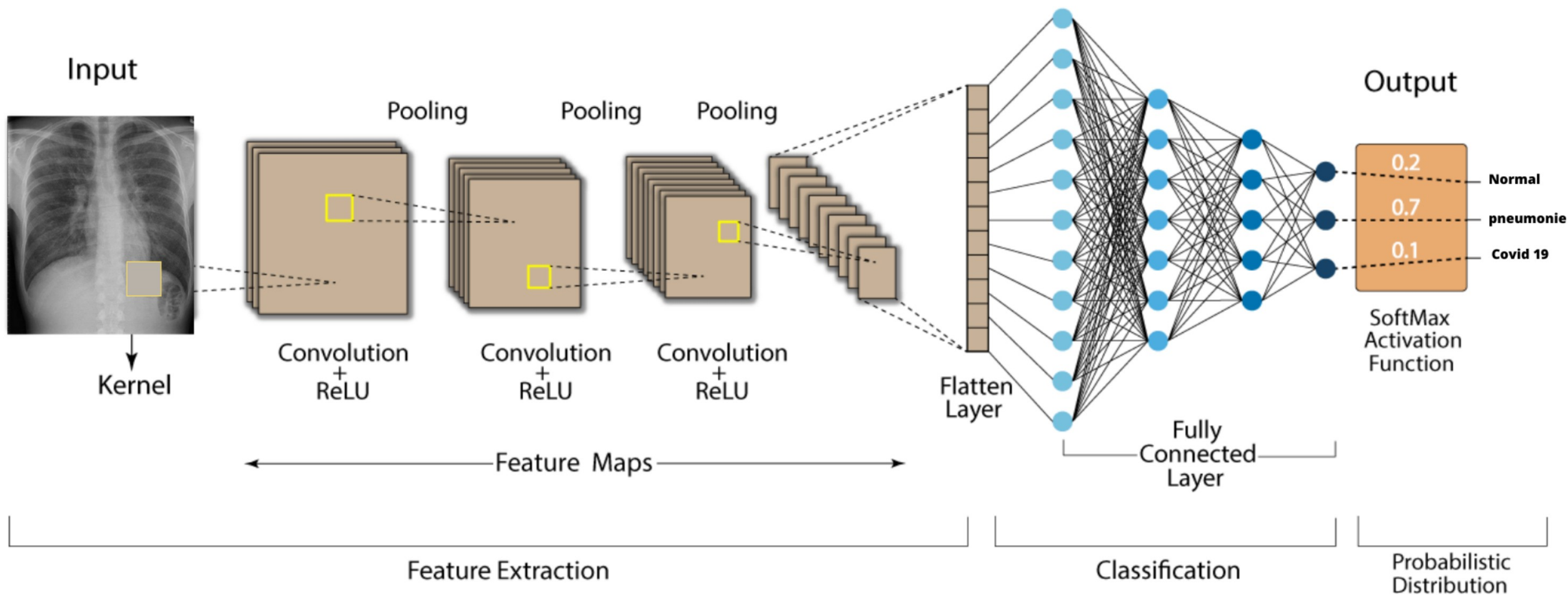
Recap

- **Spatial locality:** matrix kernel convolution
- **Translation invariance:** max pool/cross-channel pooling



Applied to Medical Imaging

Convolution Neural Network (CNN)



Applied to Medical Imaging

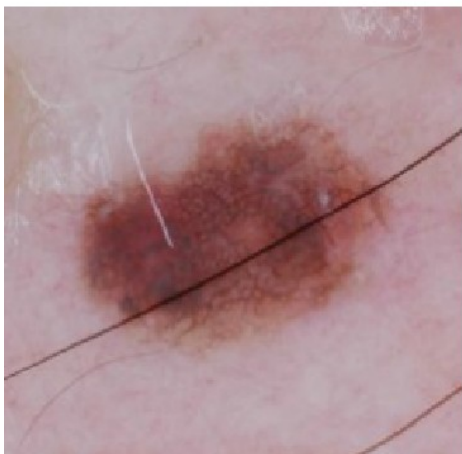
ISIC-2018^{11, 12}

ISBI-2012^{13, 14}

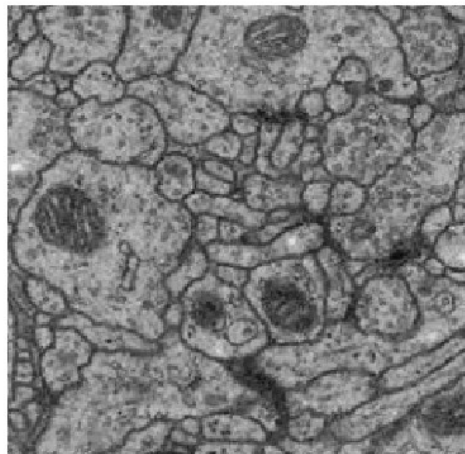
GlaS-2015¹⁶

MRI¹⁵

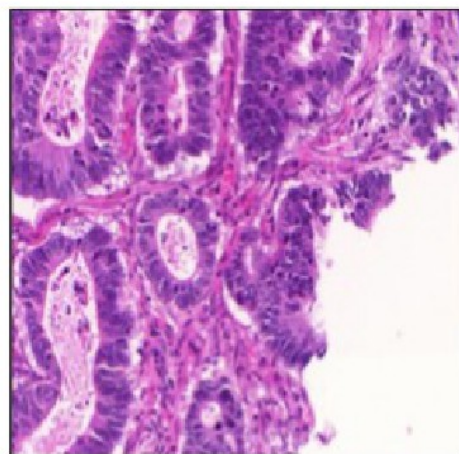
DSB-2018¹⁷



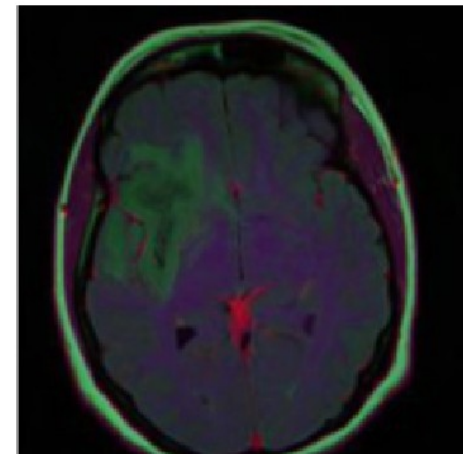
(a)



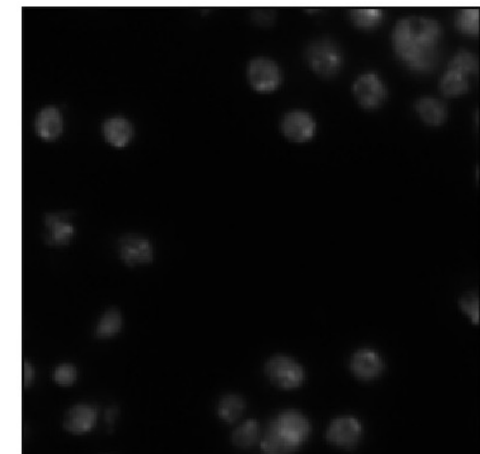
(b)



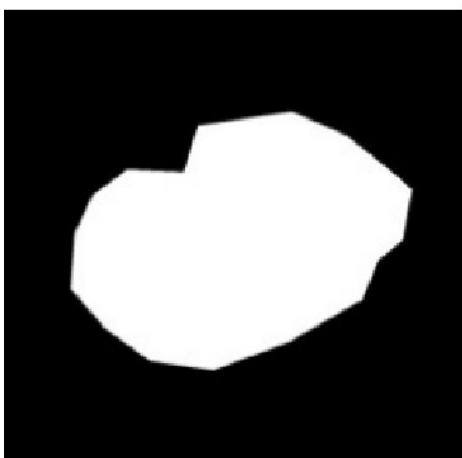
(c)



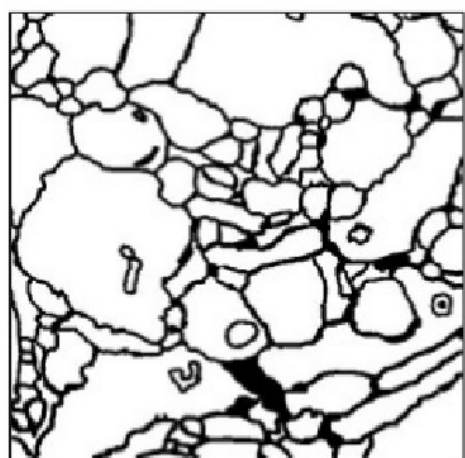
(d)



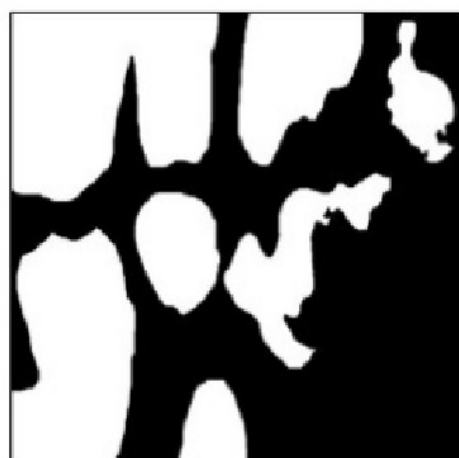
(e)



(f)



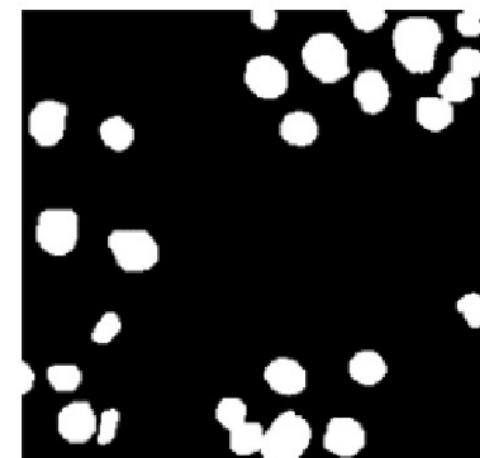
(g)



(h)

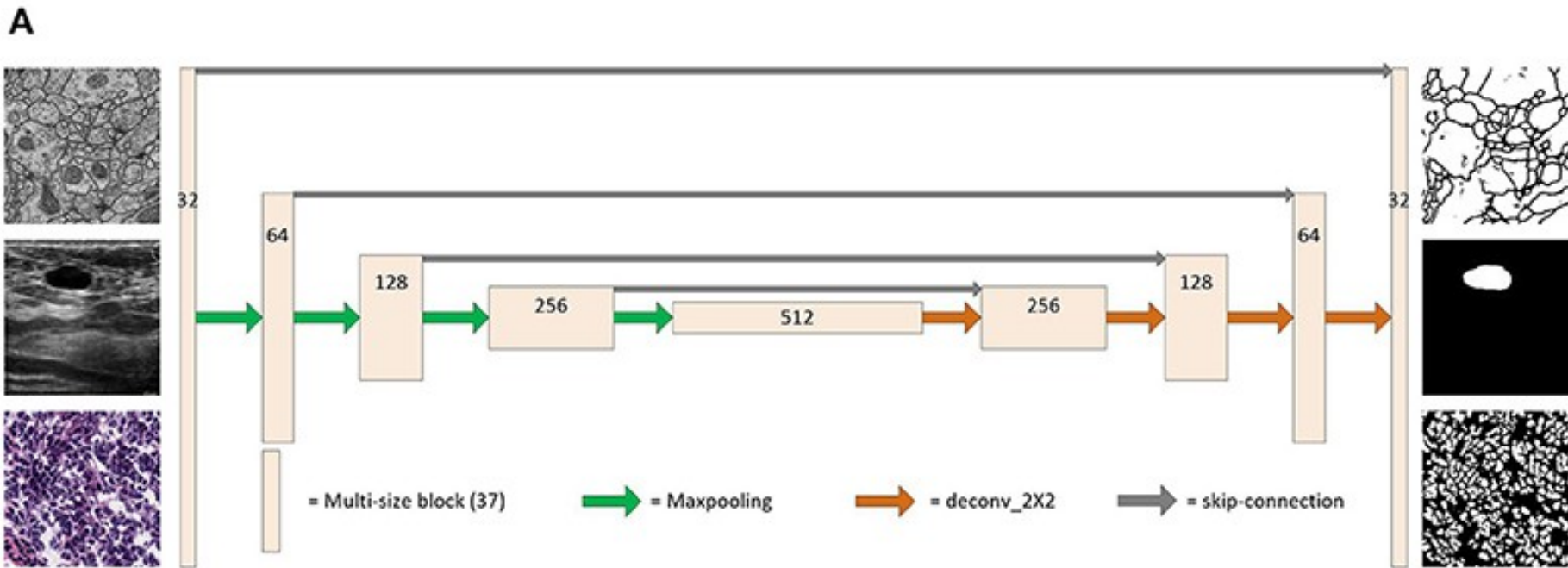


(i)

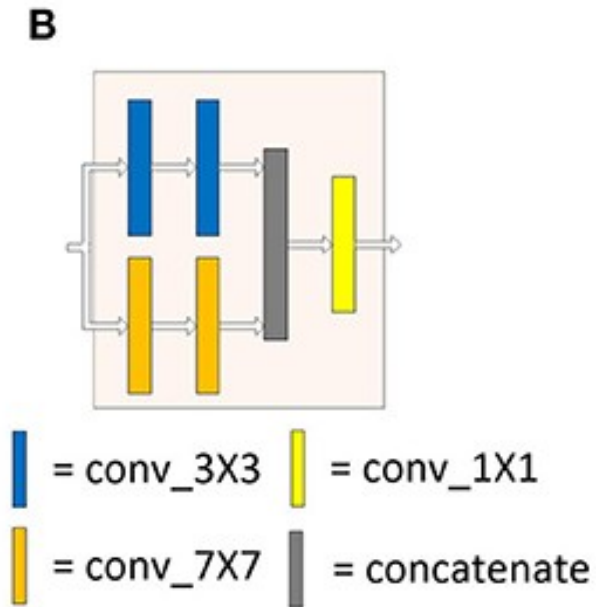


(j)

U-Net



MSU-Net

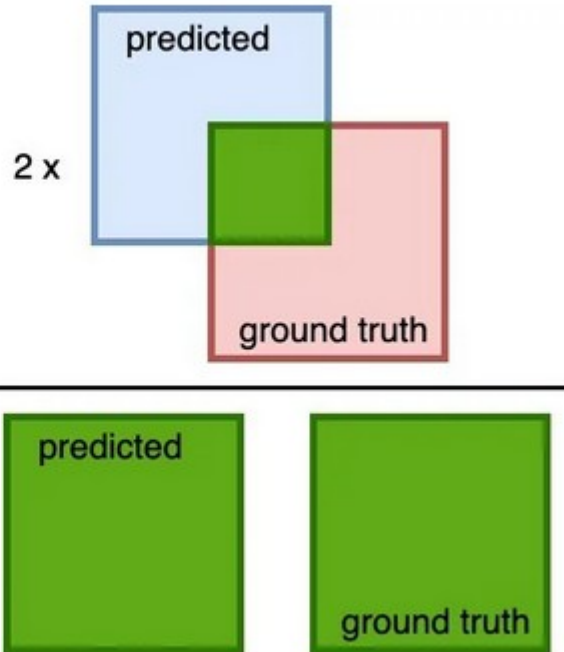


Multi-scale block (37)

- CNN variant which extracts features, then reapplies them to form a segmentation mask
- Skip connections fight loss of high-level features during max pool

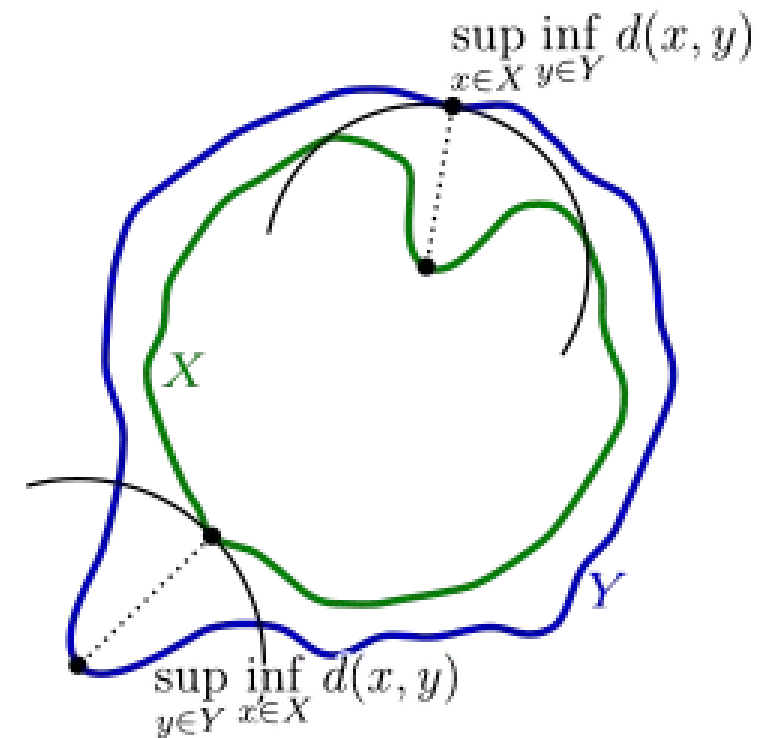
Segmentation Accuracy Metrics

Dice coefficient = $\frac{2 \times \text{area of overlapped (green)}}{\text{total area (green)}} =$



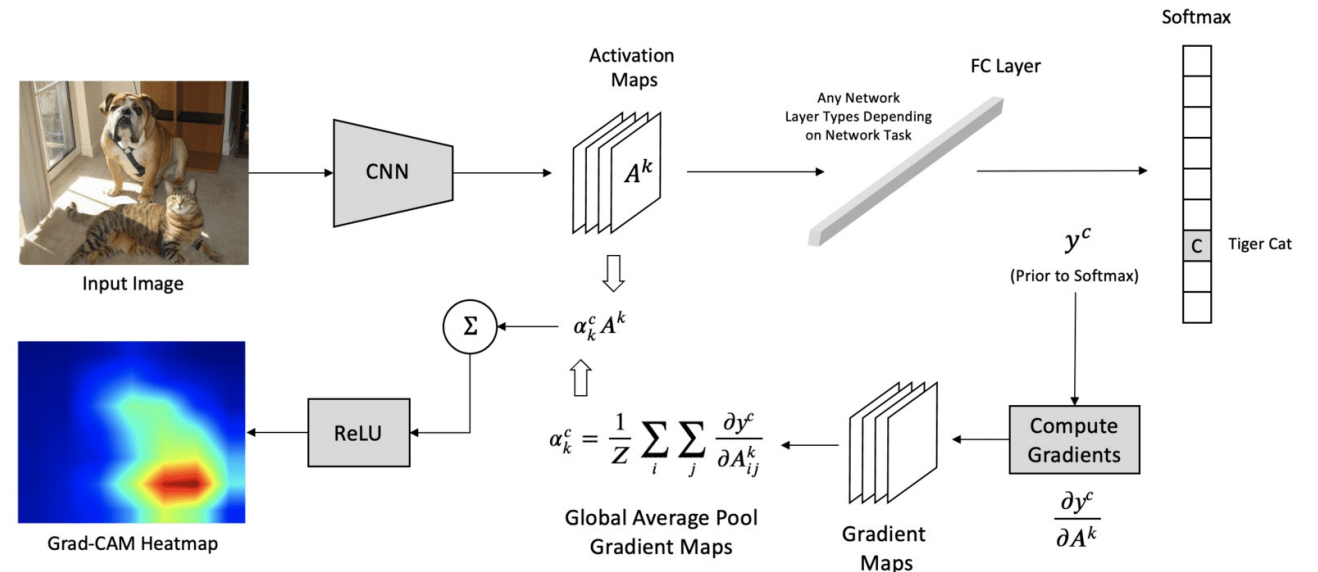
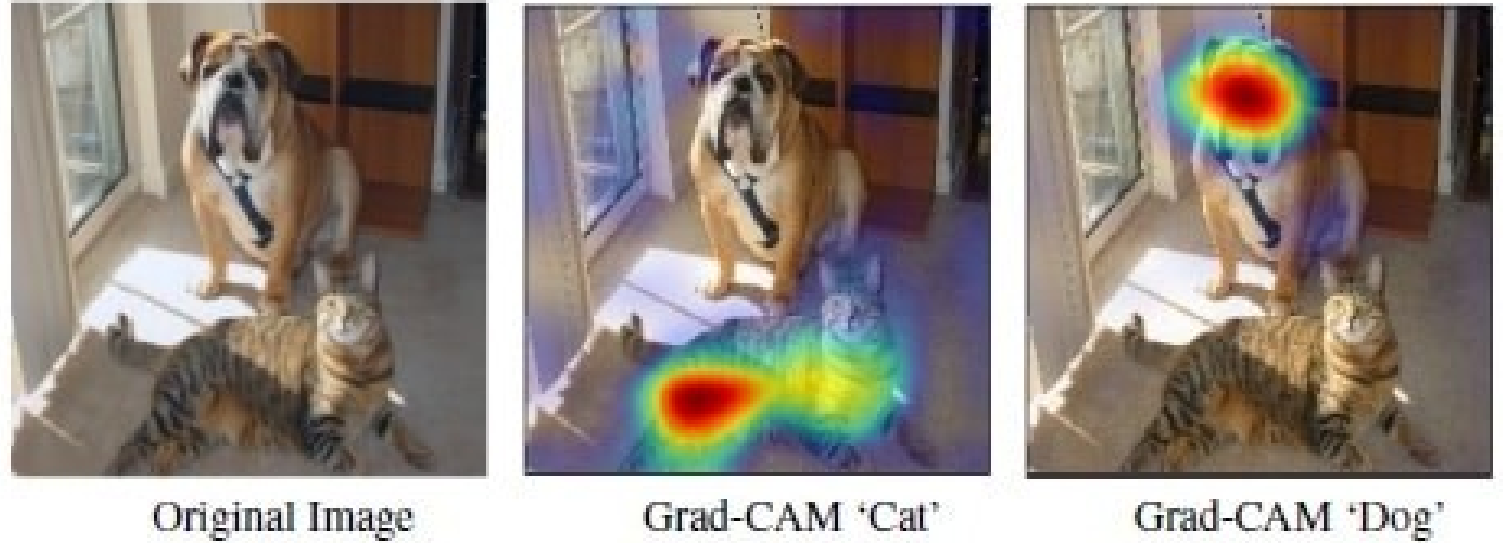
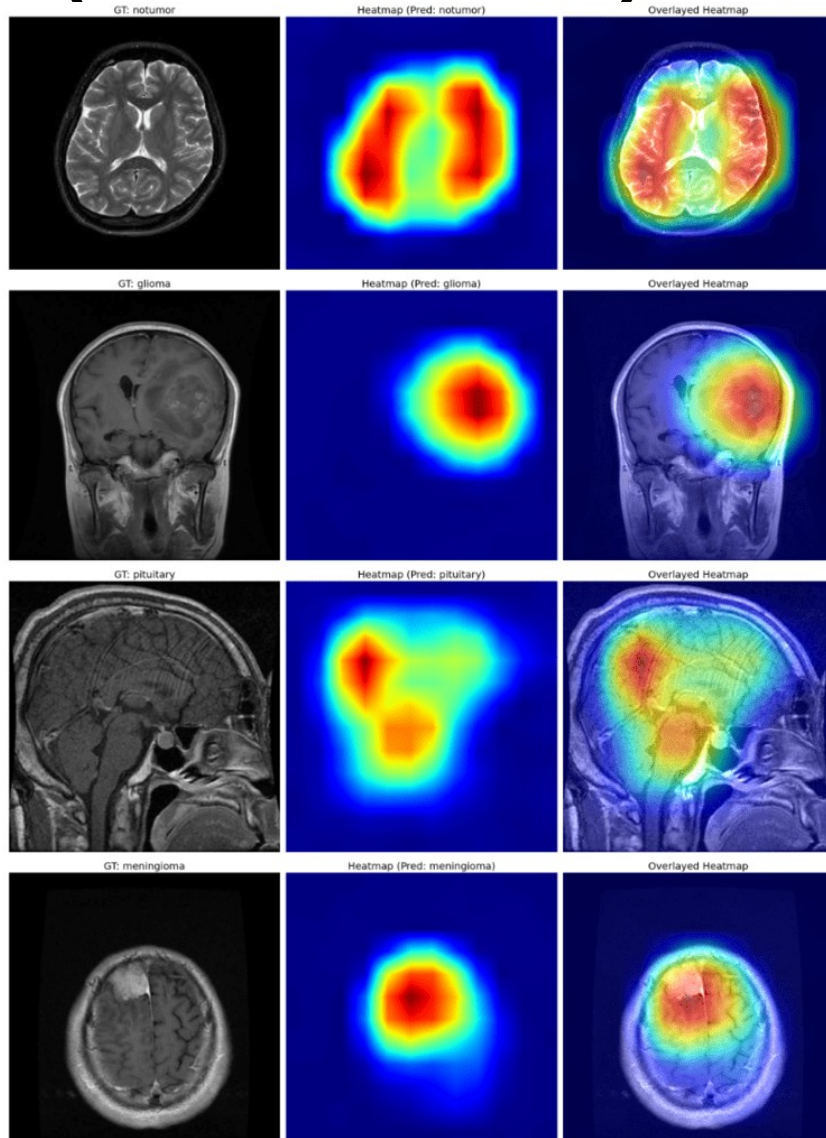
The diagram illustrates the Dice Score calculation. It shows a blue rectangle labeled 'predicted' and a red rectangle labeled 'ground truth' overlapping in a green area. Below, two green rectangles represent the 'predicted' and 'ground truth' areas used in the denominator. A '2 x' label indicates the overlap area is counted twice in the numerator.

Dice Score



Hausdorff Distance

Gradient Class Activation Mapping (GradCAM)



Radiology Workflow

Efficiency... beyond the CT room



Order entry &
protocoling

Patient
positioning

Exam setup

Scanning

Image reconstruction

Image review &
analysis



Protocols

- Given a patient's clinical indication, history, labs, previous images, allergies/contraindications, etc., decide which imaging exam + which parameters to order
 - Contrast v noncontrast
 - Contrast phase
 - MRI sequences
 - Field of view

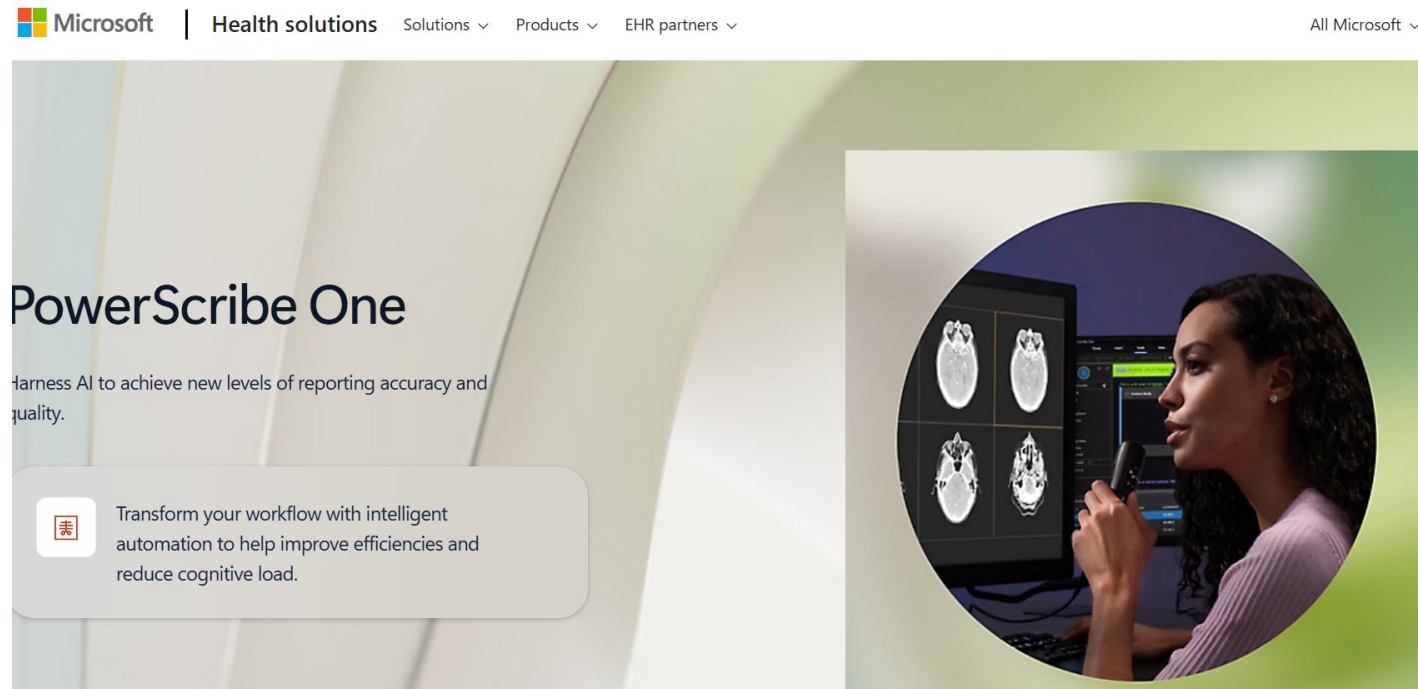
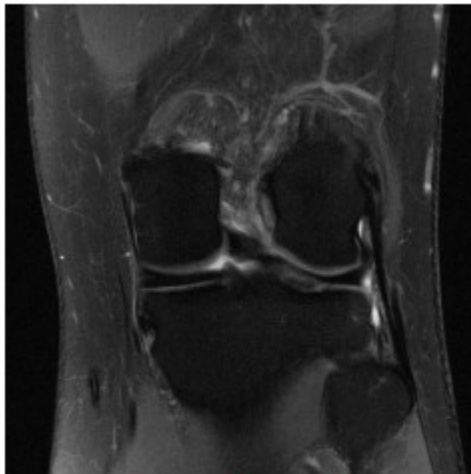
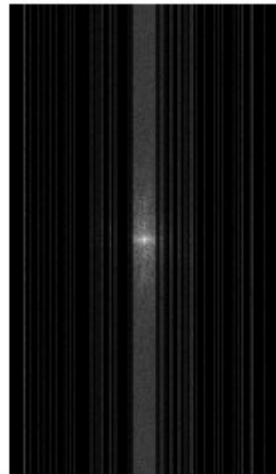


Image Acquisition/Reconstruction: FastMRI

- Reconstructs low fidelity image images using U-net
- Results in 4-8 fold speed increase in MRI acquisition



(a) Cropped and vertically flipped reconstruction from fully sampled k-space data



(b) Rectangular masked k-space



(c) Reconstruction via zero-filled IFFT (d) Deep-learning baseline UNET reconstruction

Triage and Prioritization

- Model may not make final diagnosis – but can infer diagnostic probability (which AI problem?)

- Infarct POS: AI triage score=0.68
- Mass effect POS: AI triage score=0.88
- Midline shift POS: AI triage score=0.92
- Arterial hyper-density POS: AI triage score=0.75

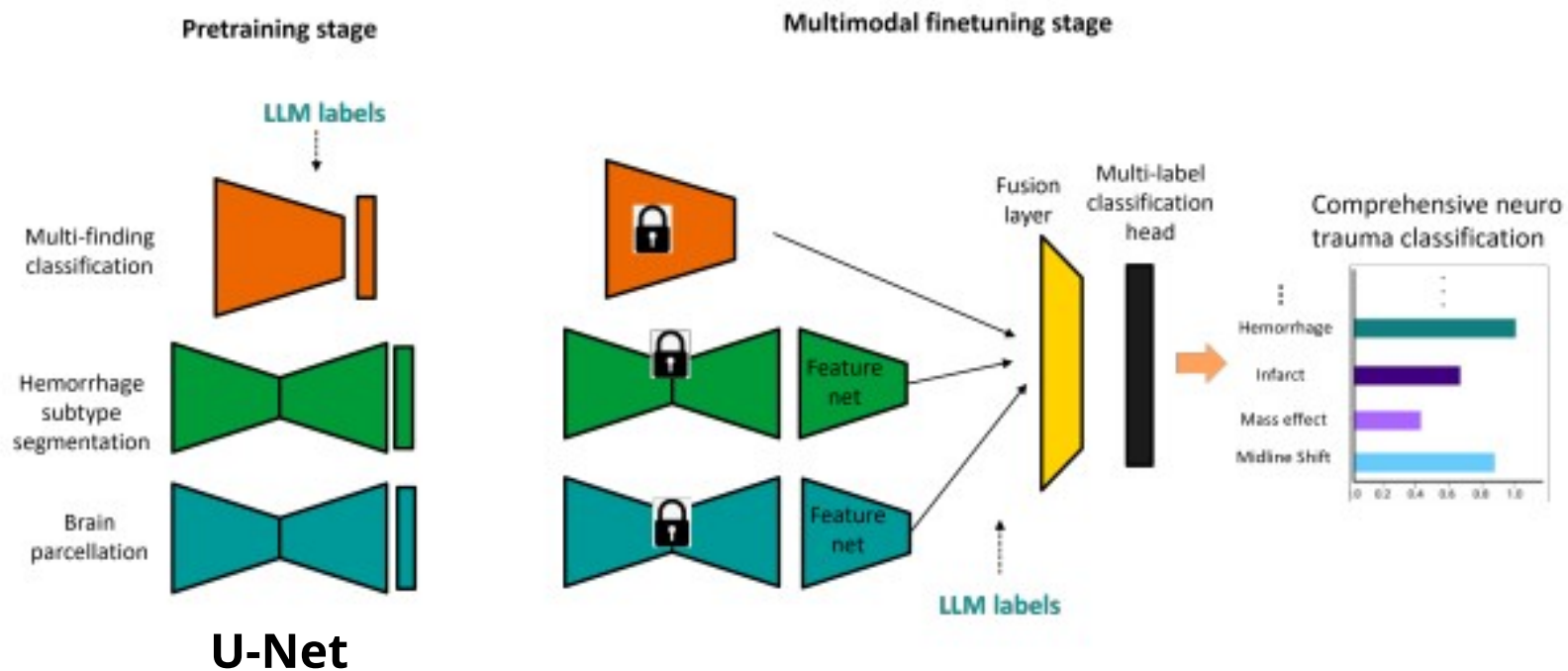
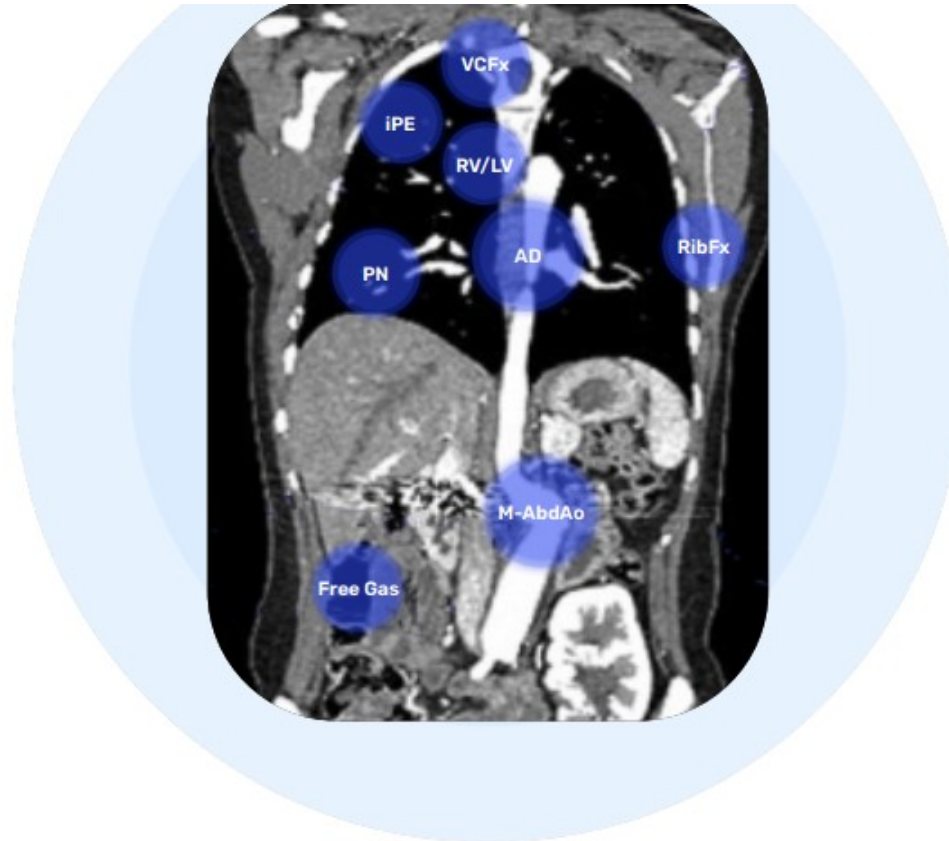


Fig. 1. Overview of the head CT foundation model for neuro-trauma triage.

Aidoc secures FDA breakthrough designation for CT triage tool

AuntMinnie.com staff writers

Sep 30, 2025



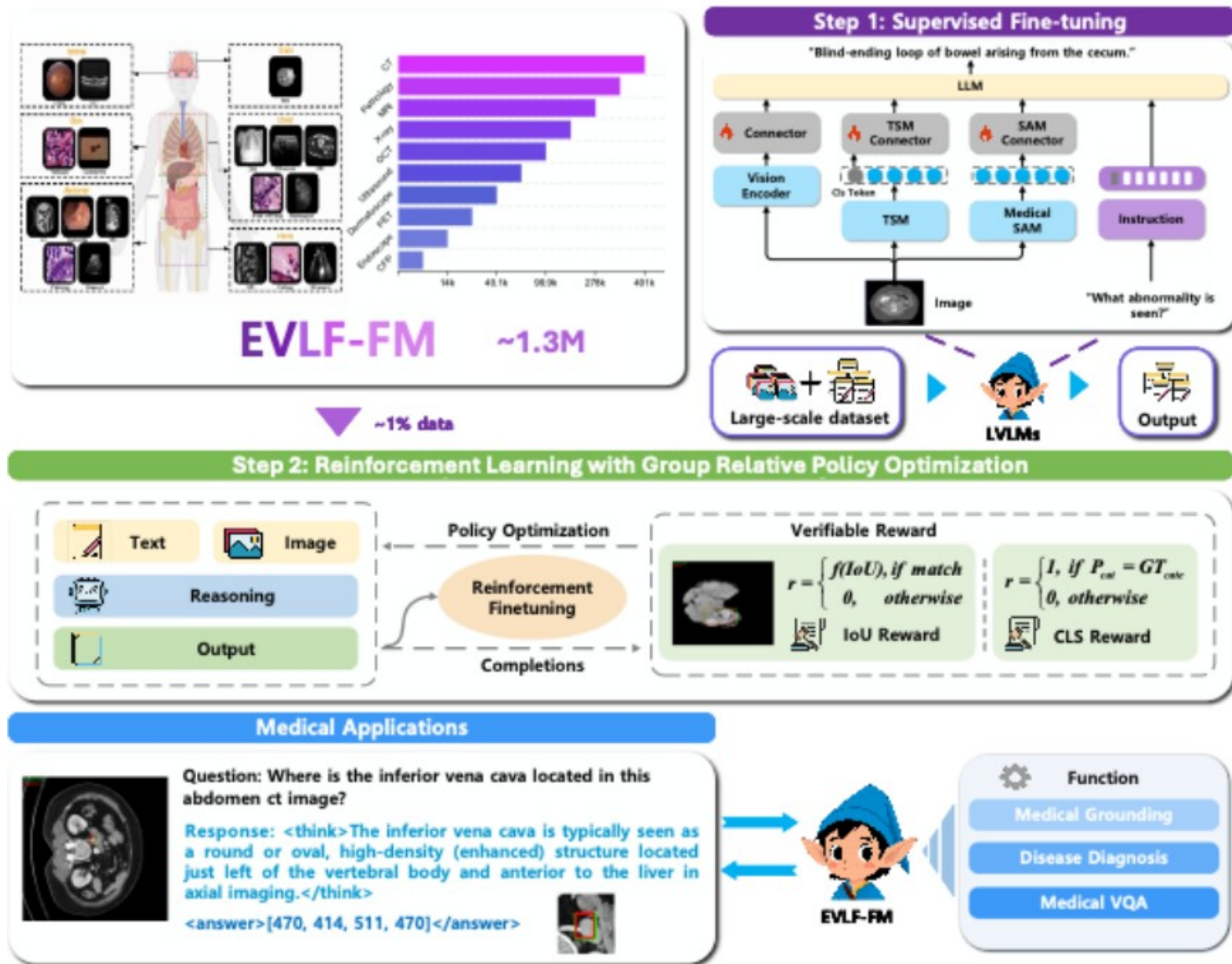
Trauma Example:

A trauma patient presented to the Emergency Department due to a car accident and a contrast enhanced CT of the chest and abdomen was ordered. Aidoc ran 8 algorithms in parallel to ensure no delay in getting the results back to the radiologist.

- Abdominal Aortic Measurements (M-AbdAo)
- Aortic Dissection (AD)
- Incidental Pulmonary Embolism (iPE)
- Intra-abdominal Free Gas
- Pulmonary Nodule (PN)
- Rib Fracture (RibFx)
- RV/LV Ratio
- Vertebral Compression Fracture (VCFx)

<https://5748396.fs1.hubspotusercontent-na1.net/hubfs/5748396/Brochures/EFM173%20-%20Radiology%20Brochure.pdf>

Multimodal – vision/text models



And Many More...

- <https://healthairegister.com/radiology/products>
- <https://arxiv.org/search/?query=radiology&searchtype=all&source=header>

New Directions

- Multimodality
- Natural Language Processing



ChatGPT



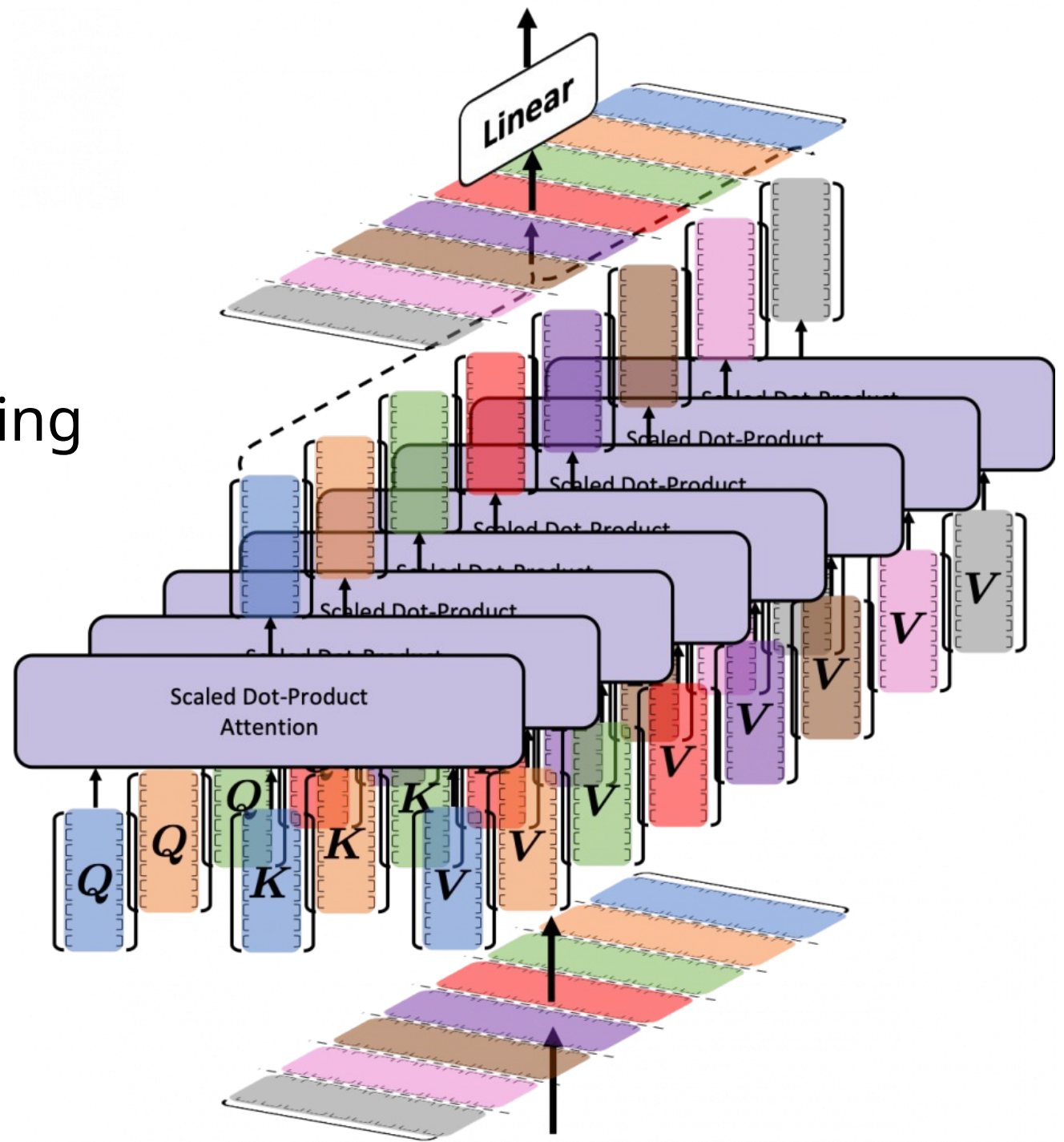
Gemini



Claude

OpenEvidence[®]

Next Time!



Thanks
!